

## 1. BIOS

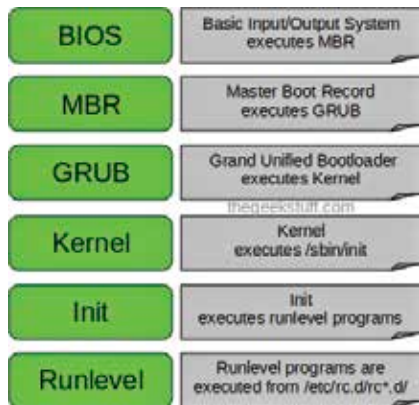
The BIOS is an acronym for Basic Input/Output System. The MBR (Master Boot Record) boot loader is loaded and executed by the BIOS. When you power on your device for the first time, the BIOS examines the HDD or SSD for integrity.

It then looks for, loads, and runs the boot loader software stored in the MBR. In certain cases, such as in a live Linux installation, the MBR is on a USB stick or CD-ROM. When the boot loader software is found, it is inserted into the memory, and the BIOS grants it control of the system.

## 2. MBR

The Master Boot Record (MBR) is in charge of loading and running the GRUB boot loader.

Depending on your hardware, the MBR is found in the first sector of the bootable disk. It is generally `/dev/hda` or `/dev/sda`. In older systems, the MBR also stores information about GRUB or LILO.



Stages of Linux boot process

## 3. GRUB

The GNU GRand Unified Bootloader, also known as GNU GRUB, is the standard boot loader for maximum current Linux computers.

When you turn on your computer, the GRUB splash screen is frequently the first thing you will see after booting Linux. It features a simple menu with a few options to choose from. If you have many different kernel images installed, then you can choose which one to boot your system with using your keyboard. The most recent kernel image is chosen by default.

The splash screen will always wait for you to pick an option for a few seconds. If you do not pick the image, the default kernel image will be loaded.

## 4. Kernel

The kernel is regarded as the heart of every operating system, including Linux. It has total command over your entire system.